Gargi Joshi

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Education

Pune Institute of Computer Technology — Pune, Maharashtra
B.Tech in Information Technology, Minor in Finance; CGPA: 9.42/10 — *Sept. 2024 – Present*

SevaSadan High School and Junior College — Nagpur, Maharashtra
Maharashtra State Board, Class XII; Grade: 87.83% — *June 2022 – May 2024*

Experience

AWS Cloud Club – PICT Pune — Aug. 2026 – Sept. 2026
Research Intern & Team Lead — *Pune, Maharashtra*

- Led a **6-member research team** on carbon-aware cloud optimization using 140,000+ Azure VM trace records.
- Built a forecasting pipeline comparing **3 models**; **Differenced Ridge** cut MAE by 10.9% vs. the baseline.
- Implemented a rule-based **scheduling prototype** that shifted 21/484 workloads to lower-carbon windows.
- Applied **INT8 quantization** to a 75K-parameter **PyTorch** model: 72.34% less memory, 42.5% lower latency.

Thyright Innovations Pvt. Ltd. — Nov. 2025 – Jan. 2026
Website Developer Intern — *Remote*

- Developed **13+ responsive website pages** with on-page **SEO** for a website serving 500+ monthly users.
- Owned **3-phase delivery** from requirements to deployment, flagging risks to ensure on-time go-live.
- Integrated **PayPal and Razorpay** payment workflows to support online transactions.
- Cut UI defects by 40% via cross-browser testing on 5+ devices and applied **DPDP Act**-compliant data handling.

Projects

Nexus Ops: Multi-Tenant Enterprise Portal — July 2026
Next.js, TypeScript, PostgreSQL, Prisma ORM, Redis, NextAuth — GitHub | Live Demo

- Built a multi-tenant enterprise portal with **25+ APIs** for support tickets, code reviews, and team management.
- Implemented **5-role RBAC** and tenant-scoped access controls to prevent cross-company data access (**BOLA**).
- Developed secure team invitations and role management, alongside **Redis-backed session revocation** across devices.
- Implemented an append-only **audit log** and wrote **10 automated test suites (172 assertions)** covering tenant isolation, authorization, and session security.

MuleRadar: AI Mule Detection Platform — Aug. 2026
Python, FastAPI, React, XGBoost, LightGBM, CatBoost, SHAP, scikit-learn — GitHub

- Built an **explainable AI platform** (**FastAPI + React**) to flag mule accounts at a **0.89%** fraud rate.
- Cut **3,861 features to 72** with a four-stage selection pipeline that removed leaked columns.
- Trained an ensemble of **XGBoost, LightGBM, CatBoost and Isolation Forest**: **0.9802 ROC-AUC**, 0.7966 PR-AUC.
- Added **SHAP** explanations and a **human-in-the-loop** audit trail so investigators can review flagged accounts.

QuantumLab: Quantum Learning Platform (SIH 2026) — 2026
Python, FastAPI, React, TypeScript, Qiskit, LLM, Three.js — GitHub | Live Demo

- Built a full-stack quantum learning platform with **FastAPI**, **React + TypeScript** and **Qiskit Aer** (up to 8 qubits).
- Created interactive **Three.js** Bloch Sphere and Q-Sphere visualizers with configurable shot sampling (128–8192 shots).
- Designed an **AI tutor** where Qiskit computes verified circuit facts before **LLM** answers, with an offline fallback tutor.
- Implemented an **adaptive assessment engine (IRT + BKT)**, JWT role-based login, and an instructor dashboard.

Technical Skills

Languages: Python, JavaScript, TypeScript, SQL
AI/ML: PyTorch, Scikit-learn, XGBoost, LightGBM, SHAP, Pandas, NumPy, RAG, LLM APIs (Groq, Gemini)
Backend & APIs: FastAPI, Node.js, Express.js, REST APIs, JWT Authentication
Frontend: React.js, Next.js, HTML5, CSS3, Tailwind CSS
Databases: PostgreSQL, MongoDB, Redis, Prisma ORM
Tools & Concepts: Git, GitHub, Vercel, Render, DSA, OOP

Achievements

- **Bank of India Hackathon 2026: Finalist**, Top 31 teams nationwide.
- **IEEE TechRush 2025, PICT Pune: 1st Place**, Frontend Web Development.
- **EY Techathon 6.0:** Shortlisted for Round 2, national-level competition.
- **PCCOE Indradhanu International Grand Challenge:** Shortlisted for Round 2.
- **GirlScript Summer of Code:** Contributed 2 pull requests merged into production repositories.